Spectral–Spatial Large Kernel Attention Network for Hyperspectral Image Classification

Chunran Wu, Lei Tong, *Member, IEEE*, Jun Zhou, *Senior Member, IEEE*, and Chuangbai Xiao

***Abstract*— Due to its ability to capture long-range dependencies, self-attention mechanism-based transformer models are introduced for hyperspectral image (HSI) classification. However, the self-attention mechanism has only spatial adaptability but ignores channel adaptability, thus cannot well extract complex spectral–spatial information in HSIs. To tackle this problem, in this article, we propose a novel spectral–spatial large kernel attention network (SSLKA) for HSI classification. SSLKA consists of two consecutive cooperative spectral–spatial attention blocks with large convolution kernels, which can efficiently extract features in spectral and spatial domains simultaneously. In each cooperative spectral–spatial attention block, we employ the spectral attention branch and the spatial attention branch to generate the attention maps, respectively, and then fuse the extracted spatial features with the spectral features. With large kernel attention (LKA), we can enhance the classification performance by fully exploiting local contextual information, capturing long-range dependencies, as well as being adaptive in the channel dimension. Experimental results on widely used benchmark datasets show that our method achieves higher classification accuracy in terms of overall accuracy (OA), average accuracy (AA), and Kappa than several state-of-the-art methods.**

***Index Terms*— Hyperspectral image (HSI) classification, large kernel attention (LKA), spectral–spatial attention.**

I. Introduction

HYPERSPECTRAL images (HSIs) usually contain hundreds of spectral channels, which include rich spatial and spectral information. Its spectral dimension is composed of tens or hundreds of sequential bands in the visible to infrared wavelength range, which has many advantages such as high spectral resolution and strong material-related feature description capability. Therefore, HSIs have been widely used in many fields such as agricultural applications [1], mineral exploration [2], environmental monitoring [3], and target detection [4], [5], [6].

In the early stage, classical machine-learning algorithms have been widely used for HSI classification [7], [8], [9], [10], [11]. However, these methods usually need to extract features designed manually, which could not achieve high classification accuracy. Over the past several years, deep-learning methods have drawn much attention in HSI classification due to their powerful feature extraction ability [12]. Chen et al. [13] first employed several convolutional and pooling layers to extract deep features from HSIs and achieved better performance than classical machine-learning methods. Among convolutional neural network (CNN)-based HSI classification methods, feature extraction and image representation can be implemented by 1-D-CNN on the spectral domain [14], 2-D-CNN on the local spatial domain [15], or 3-D-CNNs on both spectral and spatial domains [16]. Zhang et al. [17] proposed a distinct region-based CNN to encode semantic context-aware representations. Yu et al. [18] introduced an efficient CNN architecture in an end-to-end manner to improve its discriminative power for HSI classification. Chen et al. [19] proposed a migration learning-based approach that integrates deep-learning models and random subspace-based ensemble learning. Zhong et al. [20] designed a joint spectral–spatial residual network (SSRN) to extract the spectral and spatial features of HSIs, which exploits residual connection to mitigate the phenomenon of diminishing accuracy. To reduce the long training time and slow convergence of SSRN, Gao et al. [21] developed a stochastic deep residual network (SDRN) to reduce the training time by randomly dropping a layer in the training stage and restoring the entire network structure in the testing stage. However, most CNN-based deep-learning methods usually need a large number of training samples. In contrast, deep-learning methods requiring limited training samples have also been proposed, such as graph convolution network (GCN)-based methods [22], [23], [24], meta-learning-based methods [25], [26], and lightweight-based methods [27].

In recent years, the attention mechanism [28] has been introduced into HSI classification tasks, which assigns higher weights to important features and suppresses less useful features. Because CNNs cannot well mine and represent the sequential properties of spectral features, Zhang et al. [29] proposed a spectral–spatial self-attention network (SSSAN) that integrates local features with long-range dependencies. Alternative methods along this line include a cross-attentive spectral space network [30] to alleviate the HSI rotation problem and an efficient spatial attention module to reduce redundant computations in the attention mechanism [31]. Transformer, which is composed entirely of self-attention modules, has also been applied to HSI classification [32].

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Hong et al. [33] proposed SpectralFormer, which employs transformers to form a new backbone network. Following this work, the transformer-based framework has been extended to capture high-level semantic features [34] or establish dependencies between long sequence data [34]. In particular, Mei et al. [35] proposed a two-branch multiattention mechanism network (SSAN), which links the RNN-based spectral attention module and the CNN-based spatial attention module in parallel, and then combines the obtained results to extract spectral features and spatial features separately. Zhong et al. [36] designed a joint spectral–spatial transformer (SSTN) architecture to apply the self-attention mechanism for good classification accuracy with small training samples.

Despite their success, both convolution operation and self-attention have their own limitations. The convolutional operations suffer from a large number of weight parameters and a lack of global perceptibility [37]. Specifically, the number of weight parameters for convolutional operations is usually large, which increases the storage requirements of the model and requires more computational resources during training and testing. In addition, the convolutional operation is a local operation where each output pixel is only relevant to its surrounding input pixels, which may not be able to fully extract the semantic and contextual features of the image. The self-attention mechanism was originally designed for 1-D natural language processing (NLP) tasks and treats 2-D images as 1-D sequences [38]. It destroys the critical 2-D structure in images and may neglect the features between neighboring pixels. Moreover, self-attention also ignores the features in spectral dimension which is crucial information in HSIs. Recently, large kernel attention (LKA) [39] was proposed to solve these problems. LKA can be decomposed into three parts: a spatial local convolution, a spatial long-range convolution, and a channel convolution. It can extract both local and global features while preserving long-term dependencies.

In this article, we propose a spectral–spatial LKA network (SSLKA) for HSI classification. It employs LKA to extract the spectral–spatial features and local and global features while obtaining long-term dependencies. HSI cube is first preprocessed by a 2-D convolutional layer for dimension reduction. Then, the processed data is fed into the cooperative spectral–spatial LKA block to extract spectral–spatial features, which contains a spectral attention module and a spatial attention module with large kernel convolutional attention in each module. A multilayer perceptron (MLP) attention module is also attached after the spectral and spatial module to prevent the problem of degradation of feature extraction ability. After that, the spectral attention map and the spatial attention map are merged for the final classification.

Our method has several advantages over other attention-based models. Compared with SSAN [35], which embeds the attention mechanism as a module into a CNN model, SSLKA can establish global dependencies and better capture long-range dependencies. SSTN [36] first extracts spatial features and then the spectral features from HSIs, therefore, may obtain distorted features in the spectral feature extraction module. SSLKA instead adopts a parallel structure to extract the spectral–spatial features, which allows simultaneous extraction of undistorted spectral–spatial features. SSSAN [29] embeds the spectral and spatial attention modules into the Densenet, which undoubtedly increases the computational complexity. SSLKA, on the contrary, adopts the idea of a transformer and uses the entire network structure as attention modules, which makes our network relatively lightweight.

The rest of this article is organized as follows. Section II introduces the related work. Section III describes the proposed network SSLKA in detail. Section IV reports the experiments. Finally, Section V draws the conclusions.

II. Related Work

In this section, we review CNNs, attention mechanisms, and MLP. They form the basis of our method.

A. Convolutional Neural Networks

CNN is one of the representative algorithms of deep learning, which consists of various combinations of convolutional layers, maximum pooling layers, and fully connected layers. For high-dimensional feature extraction, 3-D-CNN performs well, the convolutional kernel can move along all directions (horizontal, vertical, and channel) of the image. In contrast to 2-D-CNN, which only extracts local spatial information, 3-D-CNN uniquely extracts correlation information in the channel dimension. Based on the property that 3-D-CNN can describe the spatial relationship of objects in the 3-D space, it is more widely used in HSI processing. Dilated convolution was proposed by Yu and Koltun [40] in multiscale context aggregation by dilated convolutions. It expands a convolution kernel of original size $k \times k$ to a convolution kernel of size $k + (k - 1) \times (r - 1)$, where r is the dilation rate. Dilated convolution can expand the perceptual field while keeping the size of the convolution kernel constant. To further improve the feature extraction capability, CNN-based network structures are developed toward deeper and lighter weight. Guo et al. [39] proposed the visual attention network, which decoupled the standard convolution into three parts: depthwise convolution (DW-Conv), depthwise dilation convolution (DW-D-Conv), and pointwise convolution (PW-Conv).

B. Attention Mechanism

Inspired by human vision, attention mechanisms have been actively studied in computer vision. It plays a crucial role in many tasks such as HSI classification [29], [36], target detection [4], [5], [6], and band selection [41]. The channel attention mechanism [42] takes the spatial information of each channel and compresses it to obtain a scalar representing the current channel weight. The corresponding weights of each channel are then multiplied by the feature map to obtain the weighted feature map of the channel. Not all regions of an image are equally important in contributing to the task, only the task-relevant regions are of interest. The spatial attention mechanism [43] assigns a weight to each pixel in a 2-D image, thereby enhancing or suppressing the impact of a portion of the image on the target task. Self-attention [28], derived from NLP tasks, is a specific type of attentional mechanism.

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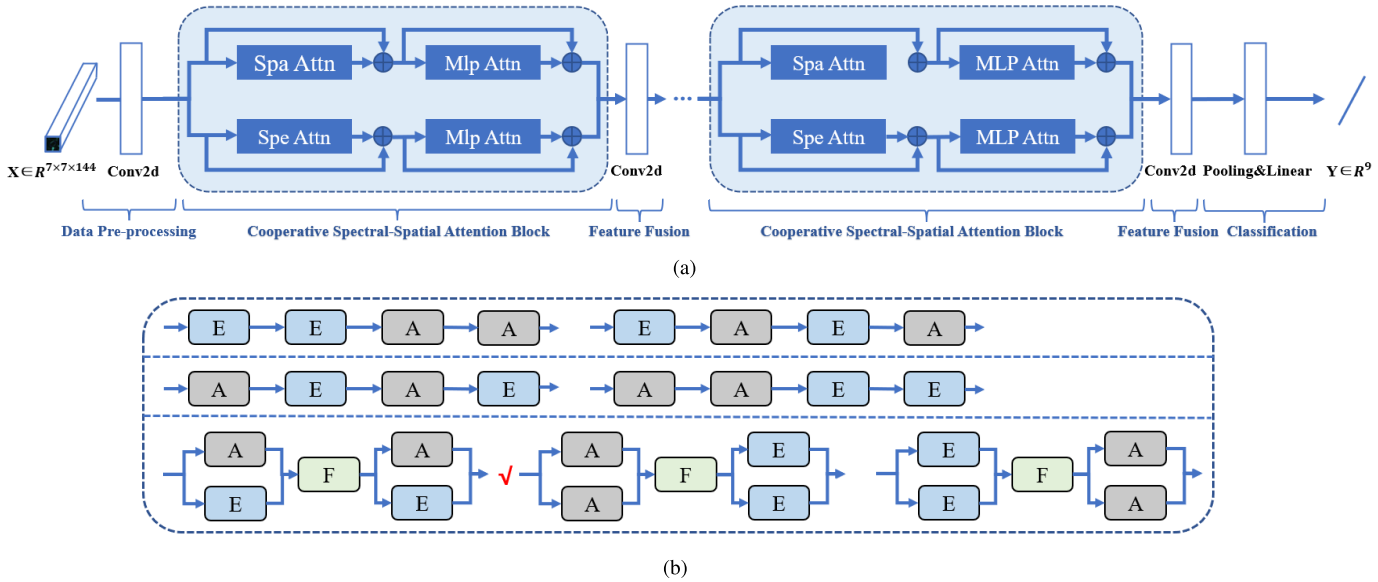


Fig. 1. (a) Illustration of the proposed SSLKA and (b) arrangement of the spectral attention branch and the spatial attention branch. (a) Structure of the SSLKA is arranged as "AE-AE," consisting of two consecutive cooperative spectral–spatial attention blocks. (b) To determine the best arrangement of the spectral attention branch and spatial attention branch, seven combinations are proposed and compared. Among them, "A" and "E" denote the spatial and spectral attention branches, respectively, and "F" denotes the feature fusion operation.

It is less dependent on external information and can better capture the internal relevance of data or features. By computing interactions between image patches, the self-attention mechanism solves the problem of long-range dependencies in image classification tasks.

C. Multilayer Perceptron

MLP [44], [45] is a widely used machine-learning tool with strong adaptive and self-learning capabilities. However, it has low learning speed and high computational requirements, which led to a long-term limitation of its capabilities. Recently, several studies have decoupled the standard MLP into a spatial MLP and a channel MLP [46]. Such decoupling reduces the computational cost and effectively reduces the parameters, resulting in a significant improvement in the performance of the MLP. For example, Guo et al. [47] proposed a MLPs with gating (gMLP), which decomposes the standard MLP and incorporates an attention mechanism. Such an approach was adopted to achieve a more accurate generation of weight maps.

III. Methodology

In this section, we describe the implementation details of the proposed method SSLKA. The network structure of SSLKA is shown in Fig. 1. HSI cube is first preprocessed by a 2-D convolutional layer. Then, the processed data is fed to the cooperative spectral–spatial LKA block, as shown in Fig. 2. It contains a spectral attention module and a spatial attention module, using large kernel convolutional attention in each module. Then, an MLP attention module is applied. After that, the spectral attention map and the spatial attention map are merged. Finally, the HSIs can be classified. To better describe SSLKA, we first introduce the LKA mechanism, which can better exploit local and global features and capture long-range dependencies. Then, we describe the cooperative spectral–spatial attention block, which is used to extract features in spectral and spatial domains simultaneously. Finally, the overall network structure is described in detail.

A. Large Kernel Attention Mechanism for HSIs

Large convolutional kernels are applied to extract the features in HSIs in our method. As shown in Fig. 3, a large convolution kernel can be decomposed into three parts: a spatial local convolution, a spatial long-range convolution, and a channel convolution. For the HSIs, the spatial local convolution and the spatial long-range convolution extract the local spatial and global spatial features. Meanwhile, the channel convolution extracts the spectral features in HSIs. Specifically, a large $K \times K$ convolution kernel can be decomposed into a $(2d-1) \times (2d-1)$ DW-D-Conv with a dilation rate d, a $\lceil (K)/(d) \rceil \times \lceil (K)/(d) \rceil$ DW-Conv and a 1×1 PW-Conv. Therefore, the dependencies between neighbor pixels and spatial long-range dependencies can be extracted by the DW-Conv and the DW-D-Conv. Meanwhile, the spectral features can be also extracted by using PW-Conv. The LKA module can be written as follows:

$$F_{\text{lka}} = \text{PW-Conv}(\text{DW-D-Conv}(\text{DW-Conv}(F))) \otimes F \quad (1)$$

where $\otimes$ denotes element-wise multiplication, F and F_{lka} represent the input data and output data, respectively. The values in the attention map indicate the importance of each feature.

B. Cooperative Spectral–Spatial Attention Block

Cooperative spectral–spatial attention blocks are proposed to extract the features in spectral and spatial domains. As shown in Fig. 2, the cooperative spectral–spatial attention block consists of a spatial attention branch and a spectral attention branch, with large kernel convolutional attention in

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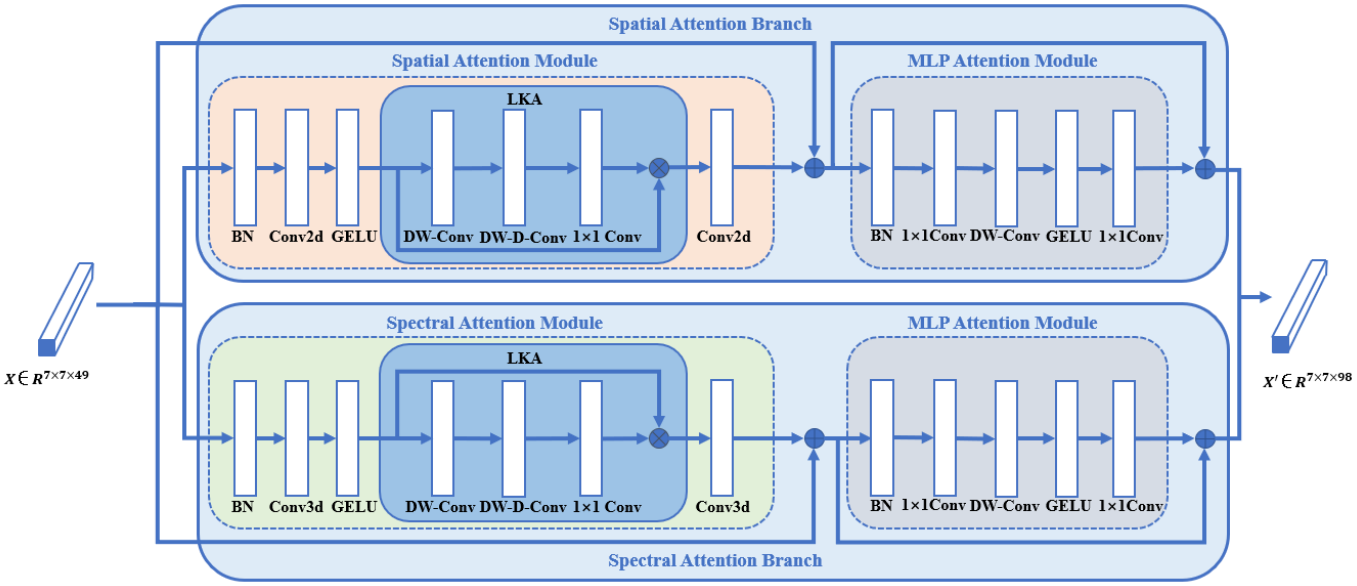


Fig. 2. Illustration of a cooperative spectral–spatial attentional block. It contains a spectral attention branch and a spatial attention branch. In each branch, an LKA module is applied first, followed by an MLP attention module.

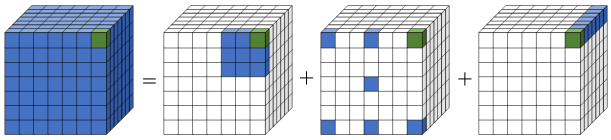


Fig. 3. Decomposition diagram of large kernel convolution. A standard convolution kernel is decomposed into a DW-Conv, a DW-D-Conv, and a PW-Conv.

each branch. Then, an MLP attention module is attached to prevent the problem of degradation of feature extraction ability. Finally, the block fusion of the features in spectral and spatial domains for final classification. Since our design of LKA is more focused on spatial feature extraction, we introduce the details of the spatial attention branch first, followed by the spectral attention branch.

1) Spatial Attention Branch: The structure of the spatial attention branch is shown in Fig. 2 (top), which is used to construct the spatial attention map. It consists of two modules: the spatial attention module and the MLP attention module. For the spatial attention module, the input features go through a batch normalization (BN) layer, a 1×1.2-D convolution layer, and a Gaussian error linear unit (GELU) to enhance the nonlinear representation, which can be formulated as follows:

$$f_{\text{pre}} = \text{GELU}(\mathcal{F}(\text{BN}(F_{\text{pro}}))) \tag{2}$$

where F_{pro} is the output of the data preprocessing. $\mathcal{F}$ denotes a 2-D convolution with the kernel size of 1×1.

Then, the LKA module, which consists of DW-Conv, DW-D-Conv, and PW-Conv, is applied to generate the spatial attention map with the local spatial, global spatial, and spectral features. After that, a 2-D convolution is also used to extract spatial features. The spatial features f_{spa} can be formulated as follows:

$$f_{\text{spa}} = \mathcal{F}(F_{\text{lka}}(f_{\text{pre}})) \oplus F_{\text{pro}} \tag{3}$$

where $\oplus$ denotes elementwise summation.

To prevent the degradation of feature extraction ability, an MLP attention module is also attached after the spectral and spatial module, which is formulated as follows:

$$F_{\text{spa}} = \mathcal{F}(\text{GELU}(\text{DW-Conv}(\mathcal{F}(\text{BN}(f_{\text{spa}}))))) \oplus f_{\text{spa}} \tag{4}$$

where F_{spa} is the output of the spatial attention branch. Shortcut connection is used in the spatial attention module and the MLP attention module to prevent the problem of degradation.

2) Spectral Attention Branch: Similar to the spatial attention branch, the structure of the spectral attention branch also consists of the spectral attention module and the MLP attention module, shown in Fig. 2 (bottom). Unlike the spatial attention module, the input feature uses the 3-D convolution with the kernel size of $1 \times 1 \times 7$ to extract the spectral information after passing through the BN layer. The GELU activation function is used to improve the generalization ability of the model, and the next step is to feed the processed tensor into the LKA module to generate a weighted feature map. At the end of the spectral attention module, the 3-D convolution is used again to extract the weighted spectral information. The above operation can be formulated as follows:

$$f'_{\text{pre}} = \text{GELU}(\mathcal{G}(\text{BN}(F_{\text{pro}}))) \tag{5}$$

$$f_{\text{spe}} = \mathcal{G}(F_{\text{lka}}(f'_{\text{pre}})) \oplus F_{\text{input}}. \tag{6}$$

Similarly, the extracted feature f_{spe} then enters the MLP attention part. Finally, F_{spe} can be formulated as follows:

$$F_{\text{spe}} = \mathcal{F}(\text{GELU}(\text{DW-Conv}(\mathcal{F}(\text{BN}(f_{\text{spe}}))))) \oplus f_{\text{spe}}. \tag{7}$$

At last, the features F_{spa} extracted from the spatial attention branch and the features F_{spe} extracted from the spectral attention branch are merged.

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TABLE I
NETWORK STRUCTURE AND PARAMETERS

Data Pre-processing	HSI Data(7,7,144)
	CONV1:1×1,49
	2×Cooperative Spectral-Spatial Attention Block
Spatial Attention	$CONV1_{spa}$:1×1,49,INPUT:CONV1 $CONV2_{spa}$:5×5,49,INPUT:$CONV1_{spa}$ $CONV3_{spa}$:5×5,49,INPUT:$CONV2_{spa}$ $CONV4_{spa}$:1×1,49,INPUT:$CONV3_{spa}$ $Multi_{spa}$,Input:$CONV1_{spa}$,$CONV4_{spa}$ $CONV5_{spa}$:1×1,49,INPUT:$Multi_{spa}$ $Add1_{spa}$,Input:CONV1,$CONV5_{spa}$
MLP Attention	$CONV1_{mlp}$:1×1,196,Input:$Add1_{spa}$ $CONV2_{mlp}$:3×3,196,Input:$CONV1_{mlp}$ $CONV3_{spe}$:5×5,49,INPUT:$CONV2_{spe}$ $Add2_{spa}$,Input:$Add1_{spa}$,$CONV3_{mlp}$
Spectral Attention	$CONV1_{spe}$:1×1×7,1,INPUT:CONV1 $CONV2_{spe}$:5×5,49,INPUT:$CONV1_{spe}$ $CONV3_{spe}$:5×5,49,INPUT:$CONV2_{spe}$ $CONV4_{spe}$:1×1,49,INPUT:$CONV3_{spe}$ $Multi_{spe}$,Input: $CONV1_{spe}$,$CONV4_{spe}$ $CONV5_{spe}$:1×1×7,1,INPUT:$Multi1_{spe}$ $Add1_{spe}$,Input:CONV1,$CONV5_{spe}$
MLP Attention	$CONV1_{mlp}$:1×1,196,Input:$Add1_{spe}$ $CONV2_{mlp}$:3×3,196,Input:$CONV1_{mlp}$ $CONV3_{mlp}$:1×1,49,Input:$CONV2_{mlp}$ $Add2_{spe}$,Input:$Add1_{spe}$,$CONV3_{mlp}$
Feature Fusion	2 Feature Fusion Module CONCAT,Input:$Add2_{spa}$,$Add2_{spe}$ CONV2:1×1,49,Input:CONCAT
Classification	Pooling,7×7,Input:CONV2 Linear:15,Input:Pooling

C. Spectral–Spatial Large Kernel Attention Network

As shown in Fig. 1, SSLKA mainly consists of several cooperative spectral–spatial attention blocks. Each block consists of a spatial and a spectral branch to extract the spatial and spectral features. In each branch, large kernel convolution is applied to generate spatial and spectral attention maps, which can extract local structural information and long-range dependencies while avoiding the drawback that self-attention ignores channel dimensionality adaptability.

Table I lists the structure and detailed network settings of the SSLKA. The input size of the data F_{input} is $c \times c \times b$, where c is the size of the patches, and b is the number of spectral bands. To eliminate redundant spectral bands and reduce the number of network parameters, a 1×1.2-D convolution is used to reduce the dimension of the spectral channels. Then, a GELU activation function is used to process the data as follows:

$$F_{\text{pre}} = \text{GELU}(\mathcal{F}(F_{\text{input}})) \tag{8}$$

where F_{pre} is the output after the preprocessing of the input data.

After data preprocessing, the output F_{pre} flows to the cooperative spectral–spatial attention blocks. In these blocks, we use the parallel connection of two attention branches to extract both spectral and spatial features. We also adapt skip connections in the spatial module, spectral module, and MLP attention module, respectively, to prevent the problem of degradation. The cooperative spectral–spatial attention block are formulated in (4) and (7). Then, the features extracted from the spatial and spectral attention branches are concatenated for feature fusion as in (9). A 1×1 2-D convolution is used for dimensionality reduction by halving the number of output channels

$$F_{\text{fus}} = \text{GELU}(\mathcal{F}(\text{Concat}(F_{\text{spa}}(F_{\text{pre}}), F_{\text{spe}}(F_{\text{pre}})))) \tag{9}$$

where F_{fus} is the final output of the feature fusion stage and Concat means concatenating the features in the spatial and spectral domains.

After the feature fusion, a pooling layer and a fully connected layer are used to classify the HSIs. The overall process of the proposed method SSLKA is shown in Algorithm 1.

Algorithm 1 Proposed SSLKA for HSI Classification

Input:
Input image;
number of iterations $\mathcal{T}$;
learning rate η;
number of cooperative spectral–spatial layers $\mathcal{L}$;
Segment the whole image into patches;
Data pre-processing of patches by Equation (8);
Train the SSLKA model;
for $t = 1$ to $\mathcal{T}$ **do**
  for $l = 1$ to $\mathcal{L}$ **do**
    Calculate the l-th layer spectral convolutional attention map F_{spe} by Equation (7);
    Calculate the l-th layer spatial convolutional attention map F_{spa} by Equation (4);
    Calculate the cooperative spectral–spatial attention map F_{fus} for the l-th layer by Equation (9);
  end for
  Calculate the gradient of the loss function in the network for each parameter by back propagation algorithm, and update the parameters with the optimization method to reduce the loss function;
end for
Conduct label prediction;
Output: Predicted label for each pixel

IV. EXPERIMENTS

To verify the effectiveness of our proposed method, we conducted a series of experiments on four widely used HSI datasets: Indian Pines (IP), Houston 2013 (HT), University of Pavia (UP), and Kennedy Space Center (KSC), respectively. In the experiments, we used overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa) to evaluate the classification performance of different methods. For a fair comparison, all the experiments were conducted ten times with the same experimental settings.

A. HSI Datasets

The IP dataset was captured by the Airborne Visual Infrared Imaging Spectrometer (AVIRS) in 1992 to image a pine forest in Indiana. It covers 16 ground classes and has a raw image size of 145×145 with 220 bands. We used 200 bands of data after excluding the absorption and low signal-to-noise ratio bands. Excluding the background elements, there are 10 249 pixels for the classification. Table II lists the number

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TABLE II
TRAINING, VALIDATION, AND TESTING DATA IN THE IP DATASET

#	Class	Training	Val	Test
1	Alfalfa	3	3	40
2	Corn-notill	55	54	1319
3	Corn-mintill	32	35	763
4	Corn	9	9	219
5	Grass-pasture	18	18	447
6	Grass-trees	28	28	674
7	Grass-pasture-mowed	3	5	20
8	Hay-windrowed	18	19	441
9	Oats	3	4	13
10	Soybean-notill	37	43	892
11	Soybean-mintill	95	87	2273
12	Soybean-clean	23	26	544
13	Wheat	8	8	189
14	Woods	49	42	1174
15	Buildings-Grass-Trees-Drives	15	15	356
16	Stone-Steel-Towers	4	4	85
-	Total	400	400	9449

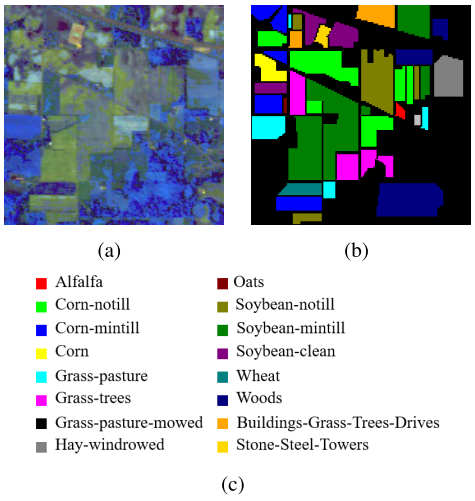


Fig. 4. IP. (a) False color image. (b) Ground truth. (c) Various classes of color legends.

of training pixels, validation pixels, and test pixels for each category. There are 16 categories in the image as shown in Fig. 4.

The HT dataset was acquired by the ITRES CASI-1500 sensor over the University of Houston campus and adjacent rural areas in Texas, USA, and has been widely used as a benchmark dataset to evaluate the performance of land cover classification. It covers 15 ground classes with an original image size of 349 × 1905. The dataset contains 144 bands and 15 029 pixels for classification. The false color image, ground-truth image, and categories of ground represented by each color in the dataset are shown in Fig. 5. Table III shows the division of the dataset.

The UP dataset was imaged in 2003 with a German airborne reflectance optical spectral imager over Pavia, Italy. It covers nine ground classes and the original image size is 610 × 340. We used 103 bands of data for classification after eliminating the noise-influenced bands, and the final number of pixels used for experimental classification was 42 776. Compared with the IP dataset, the UP dataset has a larger number of samples and fewer ground types. The specific categories of the ground in

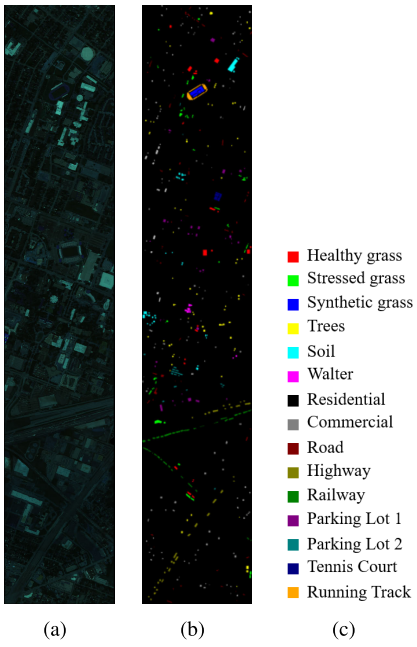


Fig. 5. HT. (a) False color image. (b) Ground truth. (c) Various classes of color legends.

TABLE III
TRAINING, VALIDATION, AND TESTING DATA IN THE HT DATASET

#	Class	Training	Val	Test
1	Healthy grass	16	32	1203
2	Stressed grass	16	32	1206
3	Synthetic grass	10	20	667
4	Trees	16	32	1196
5	Soil	16	32	1194
6	Water	6	12	307
7	Residential	16	32	1220
8	Commercial	16	32	1196
9	Road	16	32	1204
10	Highway	16	32	1179
11	Railway	16	32	1187
12	Parking Lot 1	16	32	1185
13	Parking Lot 2	7	14	448
14	Tennis Court	7	14	407
15	Running Track	10	20	630
-	Total	200	400	14429

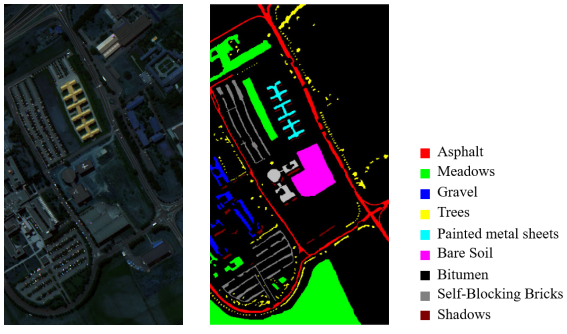


Fig. 6. UP. (a) False color image. (b) Ground truth. (c) Various classes of color legends.

the dataset are shown in Fig. 6 and the number of training, validation, and test sets are shown in Table IV.

The KSC dataset was taken by the AVIRIS sensor over the KSC in Florida on March 23, 1996. It covers 13 ground

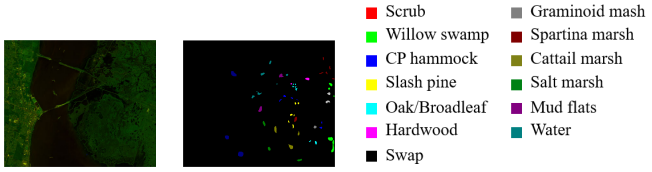
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TABLE IV
TRAINING, VALIDATION, AND TESTING DATA IN THE UP DATASET

#	Class	Training	Val	Test
1	Asphalt	31	57	6543
2	Meadows	88	163	18398
3	Gravel	10	23	2066
4	Trees	13	25	3026
5	Painted metal sheets	8	20	1317
6	Bare Soil	23	49	4957
7	Bitumen	9	19	1302
8	Self-Blocking Bricks	14	31	3637
9	Shadows	4	13	930
-	Total	200	400	42176

TABLE V
TRAINING, VALIDATION, AND TESTING DATA IN THE KSC DATASET

#	Class	Training	Val	Test
1	Scrub	29	56	676
2	Willow swamp	10	20	213
3	CP hammock	9	20	227
4	Slash pine	10	20	222
5	Oak/Broadleaf	7	16	138
6	Hardwood	9	18	202
7	Swap	6	12	87
8	Graminoid mash	16	32	383
9	Spartina marsh	19	38	463
10	Cattail marsh	15	30	359
11	Salt marsh	16	30	373
12	Mud flats	19	38	446
13	Water	35	70	822
-	Total	200	400	4611



(a) (b) (c)

Fig. 7. KSC. (a) False color image. (b) Ground truth. (c) Various classes of color legends.

classes with an original image size of 512 × 614. The dataset contains 176 bands and 5211 pixels that can be used for experiments. Table V shows the division of the number of training, validation, and test sets. Fig. 7 shows the false-color image, the ground-truth image, and the categories of the ground represented by each color for the KSC dataset.

We divided each dataset into three parts, including the training set, the validation set, and the test set. The training set was used to train the model, the validation set was used to evaluate the performance of the model in each training epoch, and the test set was used to evaluate the performance of each model. Tables II–V show the classes in each dataset and our specific divisions of the training, validation, and test sets, respectively.

B. Parameter Settings

To analyze the effect of the input size of the data cube c on the experimental results, we set the input size of data cubes as 3 × 3, 5 × 5, 7 × 7, and 9 × 9, respectively. As shown in Table VI, the cube size of 7 × 7 produces the best performance. Therefore, we set the size of the input data cube c to 7 × 7 in all other experiments.

TABLE VI
OA (%) OF THE PROPOSED METHOD WITH DIFFERENT INPUT SIZES

Spatial Szie	IP	HT	UP	KSC
3×3	90.63±1.53	88.34±1.71	92.16±2.10	93.21±1.62
5×5	93.87±1.44	90.71±1.62	95.73±1.65	95.47±0.81
7×7	95.50±0.25	**92.72±1.18**	**97.31±0.34**	**98.71±0.03**
9×9	**95.79±0.98**	91.33±1.44	96.42±0.88	97.31±0.47

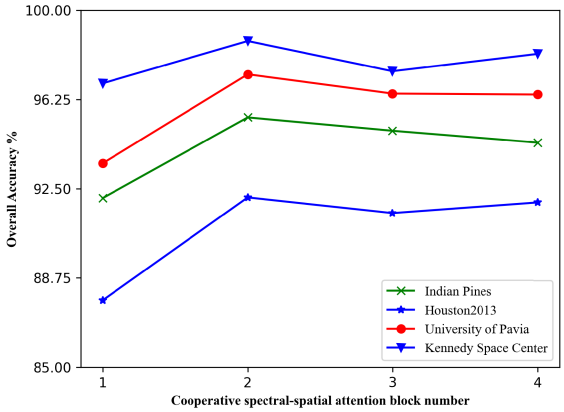


Fig. 8. OA with different numbers of cooperative spectral–spatial attention blocks on each dataset.

TABLE VII
OA OF THE NETWORK WITH DIFFERENT SIZES OF THE LARGE KERNEL IN LKA MODULE ON THE IP, HT, UP, AND KSC DATASETS

Matrices	Datasets	9× 9	11× 11	13× 13	17× 17
OA(%)	IP	94.35	94.90	**95.50**	94.36
AA(%)		94.79	95.44	**95.67**	95.06
Kappa(×100)		93.58	94.20	**94.86**	93.59
Params(K)		**130.29**	131.66	136.56	141.26
FLOPs(M)		**101.27**	102.35	106.19	109.88
OA(%)	HT	90.92	91.97	**92.14**	91.01
AA(%)		91.17	92.14	**92.72**	92.18
Kappa(×100)		90.34	91.40	**91.50**	91.39
Params(K)		**127.49**	128.86	133.76	138.47
FLOPs(M)		**99.12**	100.20	104.04	107.73
OA(%)	UP	96.39	97.13	**97.31**	96.17
AA(%)		95.60	95.52	**95.74**	94.46
Kappa(×100)		95.21	96.18	**96.43**	94.91
Params(K)		**125.19**	126.55	131.45	136.16
FLOPs(M)		**97.55**	98.62	102.46	106.15
OA(%)	KSC	97.98	98.61	**98.71**	98.48
AA(%)		96.55	97.06	**98.06**	97.67
Kappa(×100)		97.75	98.45	**98.57**	98.31
Params(K)		**126.14**	127.51	133.02	136.71
FLOPs(M)		**97.66**	99.42	103.47	106.17

To determine the optimal number of cooperative spectral–spatial attention blocks, we set their number as 1, 2, 3, and 4, respectively. As shown in Fig. 8, two consecutive cooperative spectral–spatial attention blocks achieved the best classification results, which were higher than other layer structures on OA. When the number of blocks is 1, the method fails to extract the features in spatial and spectral domains and leads to worse performance. While the number of blocks is 3 or 4, the overfitting led to lower accuracy. Therefore, we set the number of layers as 2 for the cooperative spectral–spatial attention block in all the experiments.

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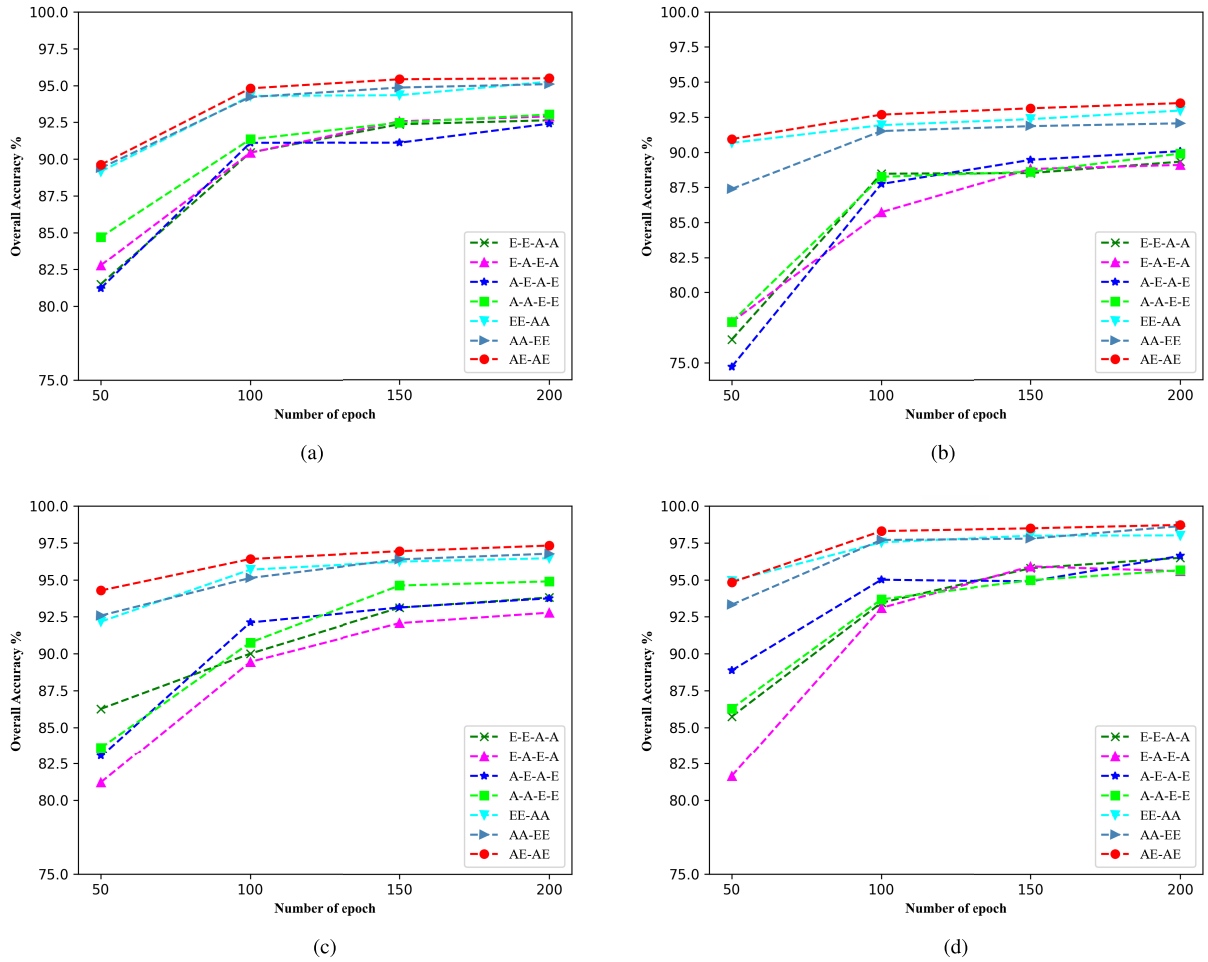


Fig. 9. OA (%) of different order structures with different training epochs. (a) IP. (b) HT. (c) UP. (d) KSC.

TABLE VIII
CLASSIFICATION RESULTS ON THE IP DATASET USING 400 TRAINING SAMPLES

Class	3D-CNN	3D-DCNN	SSRN	SDRN	SSAN	AUTO	SSTN	Proposed
1	49.22±12.56	55.33±12.30	26.48±10.37	74.85±23.90	82.43±4.05	19.47±5.44	75.20±5.86	**91.57±8.47**
2	59.30±5.90	67.93±5.37	68.88±8.63	93.61±3.25	77.15±2.97	65.17±16.82	91.95±3.67	**94.79±2.61**
3	45.94±15.71	57.10±4.71	51.20±9.57	**94.34±3.48**	75.75±2.04	67.54±8.83	92.47±4.00	93.46±6.43
4	29.89±2.48	30.90±6.06	55.04±23.11	86.31±10.73	66.67±1.90	45.29±7.86	85.85±12.48	**89.02±6.51**
5	82.39±3.60	73.38±3.95	83.23±2.54	92.45±3.91	84.03±3.61	84.01±3.52	89.28±4.20	**93.15±1.51**
6	95.15±2.16	86.99±1.40	**97.97±1.08**	99.67±0.30	90.71±2.32	98.51±1.41	98.75±0.35	99.34±0.28
7	51.17±10.83	65.00±28.28	52.95±20.17	93.53±9.34	78.95±15.78	68.72±8.94	**100.00±0.00**	**100.00±0.00**
8	92.35±5.93	92.59±2.83	97.27±2.30	**99.95±0.09**	97.64±1.18	97.77±1.80	99.92±0.11	99.63±0.73
9	97.92±2.95	84.61±10.88	60.66±18.91	97.65±4.70	81.82±0.01	88.97±13.94	**100.00±0.00**	**100.00±0.00**
10	50.39±4.21	70.70±4.10	69.10±6.54	91.43±4.16	83.04±0.12	83.25±3.18	**94.36±3.19**	90.84±3.90
11	74.64±2.83	79.26±5.88	83.59±5.89	94.12±1.32	95.83±0.53	83.18±2.93	94.80±1.25	**96.43±1.32**
12	37.02±9.08	52.14±8.85	47.20±8.86	91.37±5.61	67.56±3.82	58.07±12.39	92.20±0.31	**93.47±3.64**
13	90.35±6.52	86.42±4.58	96.68±3.75	**99.46±0.84**	88.40±4.97	97.22±2.41	99.28±0.67	98.92±1.32
14	89.72±5.95	95.71±1.38	95.47±1.67	97.84±2.04	95.53±1.34	96.47±1.44	98.21±0.46	**99.06±0.40**
15	59.14±10.85	56.93±3.91	78.43±5.88	**95.46±1.38**	71.56±7.04	50.77±9.18	92.80±3.32	91.43±3.45
16	67.05±30.14	73.33±10.27	93.18±3.84	99.54±0.92	56.80±2.47	98.12±1.58	**100.00±0.00**	99.53±0.93
OA(%)	68.71±2.16	74.46±2.12	77.67±2.77	94.63±0.70	85.67±0.34	79.41±1.47	94.48±0.38	**95.50±0.25**
AA(%)	66.98±4.12	70.40±2.26	72.33±1.38	93.85±2.28	80.87±0.96	75.16±1.11	94.07±0.31	**95.67±0.54**
Kappa(×100)	63.95±2.49	70.72±2.29	74.41±3.18	93.88±0.80	83.54±0.44	76.45±1.71	93.71±0.42	**94.86±0.29**

In our proposed SSLKA, to determine the arrangement of the spectral and spatial attention branches, seven combinations are tested. Among them, "A" and "E" denote the spatial and spectral attention branches, respectively, and "F" denotes the feature fusion operation. We tried the models with the structure of "AAEE," "AEAE," "EEAA," and so on. We conducted comparative experiments on the IP, HT, UP, and KSC datasets. As shown in Fig. 9, the structure of "AE–AE" outperforms the other structures for each dataset. This is because this structure can combine spatial attention branches and spectral attention branches in parallel and extract the spectral and spatial features more efficiently. Therefore, the structure of "AE–AE" is used in the experiments.

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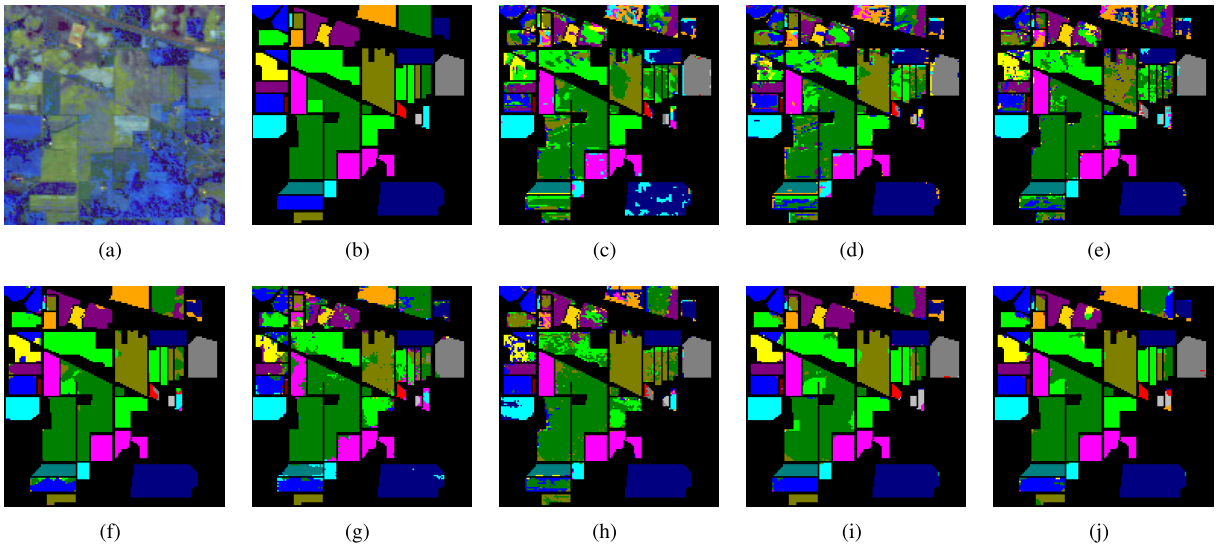


Fig. 10. Illustration of classification results on the IP dataset. (a) False color image. (b) Ground truth. (c) 3-D-CNN (69.26%). (d) 3-D-DCNN (75.17%). (e) SSRN (79.67%). (f) SDRN (94.33%). (g) SSAN (86.20%). (h) AUTO (80.14%). (i) SSTN (94.49%). (j) SSLKA (95.43%).

TABLE IX
CLASSIFICATION RESULTS ON THE HT DATASET USING 200 TRAINING SAMPLES

Class	3D-CNN	3D-DCNN	SSRN	SDRN	SSAN	AUTO	SSTN	Proposed
1	88.52±0.79	89.83±0.55	91.33±6.38	98.04±0.45	82.92±3.70	89.00±3.50	95.42±3.30	**98.59±0.47**
2	82.32±8.62	95.32±0.80	97.14±1.11	97.54±1.17	75.50±2.78	97.90±2.26	95.49±1.55	**98.06±1.23**
3	91.41±1.29	88.10±6.02	97.21±1.80	99.58±0.46	87.18±6.37	96.76±2.21	98.86±1.15	**99.95±0.07**
4	95.31±1.48	94.76±1.40	**98.82±0.80**	95.84±2.93	47.41±9.45	94.31±5.24	93.61±8.51	94.36±3.14
5	92.95±2.59	94.30±1.64	97.07±2.00	99.18±1.29	76.21±5.61	98.62±0.54	**99.85±0.16**	99.66±0.42
6	75.80±1.07	72.52±4.12	83.87±1.11	**89.08±4.98**	42.51±8.63	81.37±1.38	79.40±4.50	87.16±5.14
7	84.25±8.24	82.51±4.32	87.25±3.25	90.89±3.89	62.42±4.96	78.64±5.44	84.00±7.28	**92.67±2.42**
8	63.10±5.75	70.65±3.65	69.61±5.32	75.36±5.80	49.04±3.80	70.77±2.69	73.23±6.93	**76.52±3.51**
9	66.55±11.60	75.30±6.66	78.52±6.77	81.20±8.64	44.43±6.15	69.76±8.52	**85.58±5.58**	85.25±8.43
10	61.74±7.66	75.88±2.33	80.82±13.40	89.60±8.53	80.36±3.26	83.72±11.99	80.07±11.50	**91.88±5.22**
11	65.67±15.91	77.39±1.52	88.68±1.13	90.91±2.33	64.70±12.05	73.85±8.04	**91.65±5.62**	90.40±3.11
12	70.85±14.16	66.27±14.45	77.71±13.00	81.22±10.16	69.91±0.63	**86.96±4.07**	72.86±14.60	83.83±9.88
13	51.49±11.64	47.02±10.05	57.48±10.67	92.54±4.83	27.57±3.90	58.83±9.60	48.53±26.06	**92.59±4.54**
14	93.73±3.37	94.26±1.14	98.25±1.63	99.80±0.39	49.14±13.75	96.74±3.46	99.71±0.28	**100.00±0.00**
15	97.74±2.20	97.46±2.03	99.27±0.68	**99.90±0.12**	58.81±9.76	99.84±0.31	99.05±1.11	99.89±0.07
OA(%)	78.36±1.76	82.23±1.24	87.10±1.73	91.21±1.41	63.89±3.42	85.09±0.74	87.23±1.55	**92.14±1.69**
AA(%)	78.76±1.57	81.44±1.52	86.87±1.59	92.05±0.99	61.21±4.11	85.14±0.47	86.49±1.87	**92.72±1.18**
Kappa(×100)	76.59±1.90	80.76±1.35	86.05±1.87	90.50±1.51	60.92±3.70	83.87±0.80	86.18±1.68	**91.50±1.82**

TABLE X
CLASSIFICATION RESULTS ON THE UP DATASET USING 200 TRAINING SAMPLES

Class	3D-CNN	3D-DCNN	SSRN	SDRN	SSAN	AUTO	SSTN	Proposed
1	88.93±2.96	89.65±1.68	93.68±2.01	95.69±1.86	86.95±3.22	88.96±3.50	96.31±1.61	**97.10±0.73**
2	97.31±0.85	95.31±0.83	98.40±0.49	99.28±0.36	96.39±0.13	99.17±0.35	99.44±0.75	**99.68±0.24**
3	52.72±6.22	49.80±6.43	62.64±4.21	76.35±7.55	44.32±4.54	54.08±9.56	81.64±7.34	**85.79±2.54**
4	87.87±3.17	85.83±9.19	90.78±7.32	91.73±1.29	61.47±16.01	83.60±5.04	**97.68±2.97**	94.46±1.74
5	96.52±1.78	93.62±5.64	99.29±1.07	**100.00±0.00**	86.15±13.17	98.63±1.81	99.71±0.57	99.94±0.12
6	50.97±12.85	48.25±12.93	92.25±5.41	95.78±2.79	69.36±4.10	57.35±17.96	96.18±1.74	**96.91±1.65**
7	76.91±5.98	70.61±3.54	77.80±4.68	**99.00±0.35**	76.09±5.64	71.49±13.68	95.14±4.07	93.79±3.80
8	85.32±7.24	79.46±3.64	75.35±7.83	91.03±6.77	63.58±2.40	72.25±11.91	88.66±4.46	**95.13±3.08**
9	92.56±4.50	86.09±9.49	97.64±1.65	**99.93±0.05**	39.25±7.27	96.05±4.53	96.63±1.17	98.86±0.69
OA(%)	85.89±2.12	83.61±0.96	92.03±1.25	95.96±0.62	81.64±2.05	86.07±1.32	95.74±0.33	**97.31±0.34**
AA(%)	81.01±2.79	77.63±0.43	87.54±1.64	94.31±0.35	69.29±4.11	80.18±1.06	93.49±0.73	**95.74±0.52**
Kappa(×100)	80.92±2.95	77.82±1.35	89.41±1.68	94.63±0.82	75.12±2.96	81.07±1.94	94.33±0.44	**96.43±0.45**

The size of the large convolution kernel is also an important parameter for our proposed method. A large convolutional kernel means an expanded field of perception and more feature information can be extracted. However, as the convolution kernel enlarges, the parameters and computational cost of the model rise rapidly. Therefore, we also conducted experiments to choose the adequate size for a large convolution kernel in the LKA. We set the size of large convolutional kernels as 9×9, 11×11, 13×13, and 17×17, respectively. From Table VII, we can see that 13×13 achieved the best performance which is more suitable for our HSI classification.

C. Classification Results

To demonstrate the effectiveness of our proposed method, SSLKA is compared with seven different HSI image classification methods. It includes deep learning methods, such

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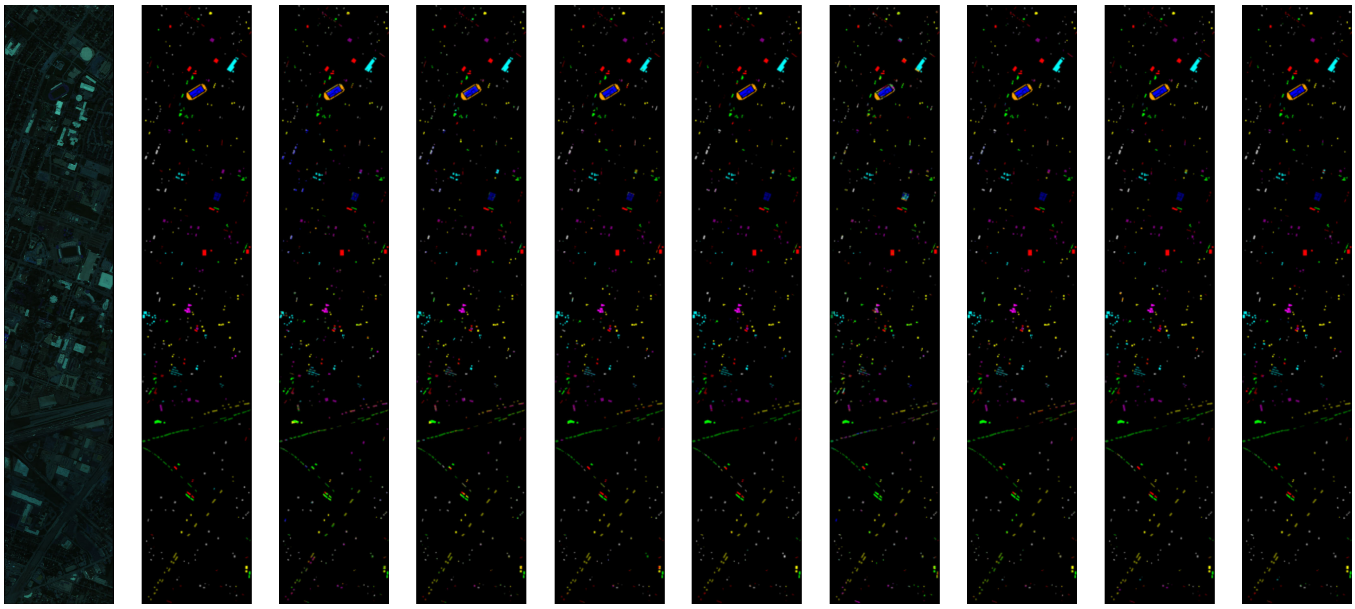


Fig. 11. Illustration of classification results on the HT dataset. (a) False color image. (b) Ground truth. (c) 3-D-CNN (79.21%). (d) 3-D-DCNN (80.88%). (e) SSRN (87.67%). (f) SDRN (90.43%). (g) SSAN (64.20%). (h) AUTO (85.14%). (i) SSTN (88.49%). (j) SSLKA (92.13%).

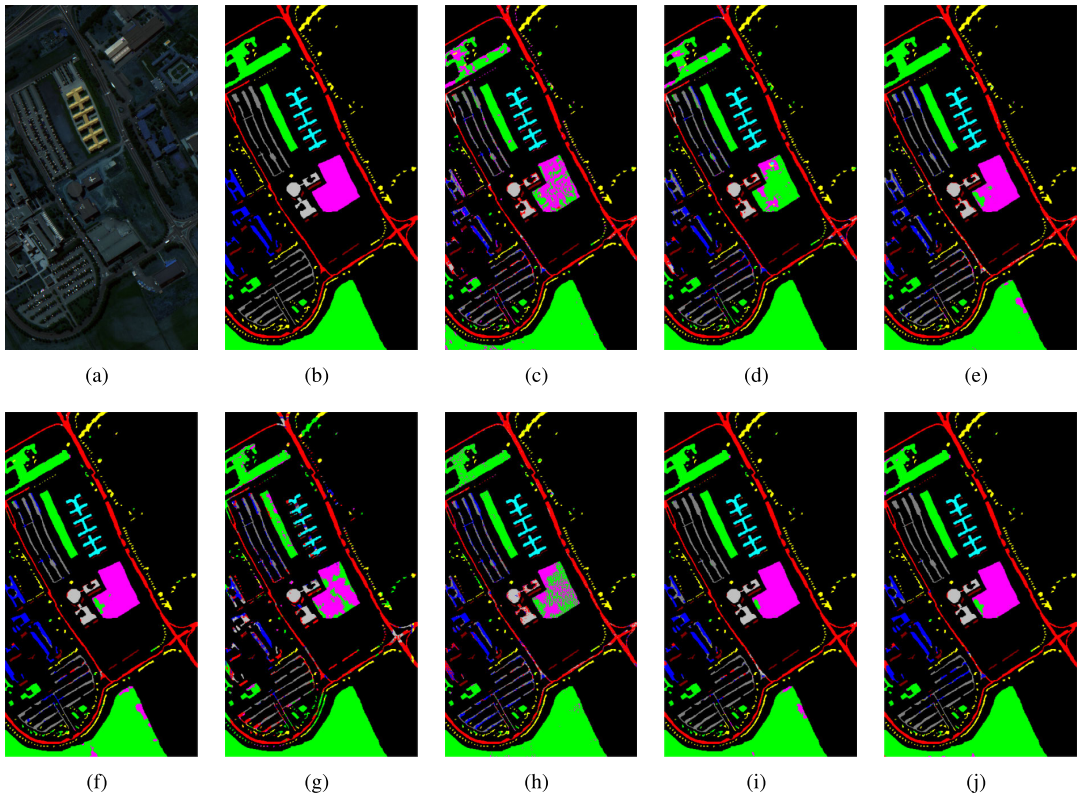


Fig. 12. Illustration of classification results on the UP dataset. (a) False color image. (b) Ground truth. (c) 3-D-CNN (86.21%). (d) 3-D-DCNN (83.54%). (e) SSRN (92.67%). (f) SDRN (95.43%). (g) SSAN (81.20%). (h) AUTO (86.14%). (i) SSTN (95.49%). (j) SSLKA (97.13%).

as 3-D-CNN [13], 3-D-DCNN [48] (which based on 3-D convolution but with smaller kernel size and deeper layers), automatic design of convolutional neural network for hyperspectral image classification (AUTO) [49], SSRN [20], and SDRN [21], with methods involving attention mechanisms like SSAN [35] and SSTN [36]. All classification methods were run in the same experimental environment, with 200 epochs trained each time, and the experiments were repeated ten times to calculate the average OA, AA, and kappa.

1) Results on the Indian Pines Dataset: Table VIII lists the results of our proposed method and the other methods on the IP dataset, where the best results are shown in bold.

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TABLE XI
CLASSIFICATION RESULTS ON THE KSC DATASET USING 200 TRAINING SAMPLES

Class	3D-CNN	3D-DCNN	SSRN	SDRN	SSAN	AUTO	SSTN	SSLKA
1	93.74±1.33	95.17±0.50	99.51±0.42	96.45±1.48	99.35±0.54	**99.95±0.06**	99.80±0.18	99.41±0.84
2	69.17±2.77	81.69±2.13	85.13±3.99	93.43±3.66	82.47±8.22	87.32±6.27	91.39±4.24	**96.87±4.10**
3	87.96±3.22	82.08±12.38	90.90±2.99	94.71±5.32	89.72±5.34	85.31±3.78	93.54±5.24	**97.50±0.55**
4	33.18±9.83	55.55±6.42	49.55±21.77	82.43±4.95	45.34±4.67	53.75±7.75	35.14±7.24	**94.59±2.05**
5	52.66±8.98	63.04±20.41	59.90±7.16	92.75±4.84	70.29±3.55	58.21±12.98	71.01±11.81	**95.41±2.39**
6	63.37±19.25	56.93±17.48	80.69±5.25	96.70±3.62	76.07±7.50	62.87±22.25	91.91±4.45	**98.18±1.30**
7	88.51±9.89	95.78±3.29	87.74±9.12	**98.08±2.71**	96.93±3.55	97.70±2.48	85.44±18.96	97.70±3.25
8	64.14±9.71	64.58±12.38	79.20±3.39	98.61±0.25	91.55±3.82	92.95±6.57	81.90±20.28	**98.69±0.93**
9	84.59±4.51	92.66±5.33	97.77±2.43	**100.00±0.00**	99.93±0.10	99.93±0.10	95.68±6.11	**100.00±0.00**
10	79.94±6.12	76.78±3.08	95.17±2.64	99.44±0.39	97.21±2.56	97.31±2.28	88.49±1.48	**100.00±0.00**
11	98.30±1.55	97.41±1.49	98.93±0.79	**100.00±0.00**	99.01±0.67	99.73±0.22	96.78±1.53	97.41±2.94
12	97.38±0.42	96.78±0.38	95.07±3.66	98.28±1.58	95.14±2.67	94.69±3.91	87.14±5.41	**99.03±1.37**
13	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.96±0.06	99.96±0.05	**100.00±0.00**	**100.00±0.00**
OA(%)	84.13±0.52	86.34±2.98	91.19±1.02	97.35±0.67	92.48±0.08	92.19±1.03	90.30±0.83	**98.71±0.03**
AA(%)	77.92±1.02	81.42±4.49	86.12±1.76	96.22±1.00	87.92±0.47	86.90±1.97	86.02±0.25	**98.06±0.27**
KAPPA(×100)	82.32±0.57	84.78±3.34	90.18±1.14	97.05±0.75	91.62±0.09	91.29±1.17	89.19±0.92	**98.57±0.03**

TABLE XII
OA OF EACH METHOD WITH DIFFERENT SIZES OF TRAINING SAMPLES ON IP, HT, UP, AND KSC DATASETS

Samples	Datasets	3D-CNN	3D-DCNN	SSRN	SDRN	SSAN	AUTO	SSTN	SSLKA
100	IP	53.03±1.51	59.77±0.94	59.23±2.10	80.76±0.73	69.24±1.56	63.21±0.52	81.17±0.56	**80.29±2.50**
200		59.00±3.53	64.48±0.62	65.50±1.95	90.11±0.35	74.97±1.37	71.48±1.21	91.37±0.30	**91.95±0.94**
300		66.53±3.36	71.25±1.47	73.60±2.11	93.71±0.41	83.37±1.13	77.57±1.28	93.85±0.53	**94.07±0.84**
400		68.71±2.16	74.46±2.12	77.67±2.77	94.63±0.70	85.67±0.34	79.41±1.47	94.48±0.38	**95.50±0.25**
100	HT	67.54±4.81	74.81±1.2	76.30±1.36	79.84±1.02	59.03±2.35	78.68±0.66	77.16±2.01	**84.84±2.13**
200		78.36±1.76	82.23±1.24	87.10±1.73	91.21±1.41	63.89±3.42	85.09±0.74	87.23±1.55	**92.14±1.69**
300		78.99±3.25	85.14±0.75	89.32±1.98	91.58±1.01	66.57±1.43	87.15±0.95	90.25±1.05	**94.13±0.97**
400		84.79±0.51	87.12±1.64	92.15±0.81	92.65±0.75	67.84±1.16	90.65±0.67	92.52±1.02	**94.76±0.48**
100	UP	79.25±1.40	77.98±1.23	84.90±0.64	90.14±0.72	77.76±1.57	81.19±1.07	91.49±0.67	**93.56±0.44**
200		85.89±2.12	83.61±0.96	92.03±1.25	95.96±0.62	81.64±2.05	86.07±1.32	95.74±0.33	**97.31±0.34**
300		86.30±2.15	86.38±1.71	93.74±0.76	97.22±0.45	83.41±2.10	89.28±0.03	97.18±0.26	**97.38±0.69**
400		91.40±1.30	89.16±1.36	94.98±1.03	97.58±0.31	85.55±1.18	91.84±0.83	97.57±0.40	**98.43±0.46**
100	KSC	78.71±3.40	78.75±1.02	82.98±2.42	94.88±0.54	84.92±0.98	84.47±84.47	86.09±0.54	**95.27±0.71**
200		84.13±0.52	86.34±2.98	91.19±1.02	97.35±0.67	92.48±0.08	92.19±1.03	90.30±0.83	**98.71±0.03**
300		88.69±0.61	89.72±2.42	91.27±0.84	97.98±0.58	93.75±0.84	93.35±0.49	90.67±3.33	**98.77±0.16**
400		90.05±1.99	93.09±0.81	94.48±1.06	98.33±0.24	94.97±0.59	94.32±0.60	91.56±1.65	**98.95±0.26**

TABLE XIII
TRAINING AND TESTING TIME ON FOUR DATASETS

		IP	HT	UP	KSC
3D-CNN	Training(s)	586.02	298.66	186.66	311.75
	Testing(s)	0.93	1.04	1.93	0.78
3D-DCNN	Training(s)	107.45	106.74	113.48	106.24
	Testing(s)	0.32	1.15	1.61	0.49
SSRN	Training(s)	579.19	246.08	188.35	303.21
	Testing(s)	8.37	9.27	19.92	17.83
SDRN	Training(s)	124.87	70.25	57.08	72.81
	Testing(s)	1.50	1.69	3.44	0.64
SSAN	Training(s)	1828.97	1692.48	1534.64	1568.49
	Testing(s)	65.82	79.63	165.02	68.37
AUTO	Training(s)	265.67	162.12	162.91	179.58
	Testing(s)	0.93	1.32	3.70	1.02
SSTN	Training(s)	114.02	75.75	75.38	73.81
	Testing(s)	0.62	0.87	2.32	0.30
SSLKA	Training(s)	156.76	96.33	98.23	101.14
	Testing(s)	0.60	0.84	2.24	0.31

Table VIII shows that the classification performance of the attention-based approach is better than that of the traditional deep-learning approaches 3-D-CNN and 3-D-DCNN. Compared with the SSTN using the self-attention mechanism, our method outperforms it in terms of OA, AA, and Kappa, which indicates that our proposed method works better in feature extraction. Fig. 10 shows the visualization results of the SSLKA method and other methods. From Fig. 10, it can be seen that our proposed method can get better results than other methods.

2) Results on the University of Houston Dataset: Table IX shows the results on the HT dataset. Our method greatly improves the performance compared to other methods. Based on LKA, SSLKA outperforms the SDRN, the second highest classification accuracy, by 0.93% and by 4.91% over the SSTN method. SDRN employs ResNet-based shortcut connections as a way to directly add shallow inputs directly to deep outputs, allowing gradients to propagate more easily to the shallow layers, thus helping to solve the gradient vanishing problem. Our method also takes advantage of ResNet by using short connections to prevent gradient vanishing. More importantly, our method incorporates features from both spectral and spatial dimensions to better learn hard-to-classify sample points through a parallel branching structure. From Fig. 11, we can see that compared with other methods, our method has better classification results and smoother visual effects.

3) Results on the University of Pavia Dataset: Table X demonstrates the results on the UP dataset. As seen from Table X, our proposed SSLKA has a great advantage in various metrics with a relatively low standard deviation. Among them, the accuracy of OA is 1.57% higher than that of the SSTN model, which fully indicates that LKA has better adaptability in the channel dimension compared to self-attention. Fig. 12 shows the visualization results of our proposed method SSLKA and other methods. It can be seen from Fig. 12 that

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TABLE XIV
ABLATION STUDY OF DIFFERENT MODULES IN SSLKA

Matrices	Datasets	w/o DW-Conv	w/o DW-D-Conv	w/o PW-Conv	w/o Spa-Branch	w/o Spe-Branch	w/o LKA	SSLKA
OA(%)	IP	94.38	95.03	94.81	91.57	91.21	93.95	**95.50**
AA(%)		93.55	94.66	95.12	92.72	88.08	92.96	**95.67**
Kappa(×100)		93.59	94.33	94.08	90.42	89.95	93.11	**94.86**
Params(K)		131.46	131.46	126.76	**64.04**	73.61	116.57	136.56
FLOPs(M)		102.35	102.35	98.66	**43.85**	56.33	90.98	106.19
OA(%)	HT	90.65	91.02	90.67	87.46	87.79	89.88	**92.14**
AA(%)		90.96	91.48	91.05	88.26	88.75	90.82	**92.72**
Kappa(×100)		89.89	90.29	89.92	86.44	86.82	89.06	**91.50**
Params(K)		128.67	128.67	123.96	**61.24**	70.82	113.77	133.76
FLOPs(M)		100.20	100.20	96.51	**40.11**	54.17	88.83	104.04
OA(%)	UP	96.94	95.96	96.61	89.49	93.31	95.18	**97.31**
AA(%)		95.70	94.17	95.04	87.55	90.18	94.87	**95.74**
Kappa(×100)		95.95	94.65	95.49	86.17	91.11	93.41	**96.43**
Params(K)		126.36	126.36	121.65	**58.93**	68.70	111.46	131.45
FLOPs(M)		98.62	98.62	94.93	**42.53**	52.59	87.25	102.46
OA(%)	KSC	96.79	97.89	97.98	92.53	93.25	96.15	**98.71**
AA(%)		94.66	96.52	96.94	89.07	91.95	94.17	**98.06**
Kappa(×100)		96.42	97.65	97.75	92.80	92.7	95.46	**98.57**
Params(K)		128.22	128.49	123.01	**62.71**	72.28	113.47	133.02
FLOPs(M)		99.36	98.97	95.74	**41.34**	55.4	86.91	103.47

our method SSLKA can handle the boundary information better.

4) Results on the Kennedy Space Center Dataset: Table XI lists the results on the KSC dataset. Our accuracy is also better than other methods, due to effectively extracting the features using the cooperative spectral–spatial attention block. For some categories, such as spartina marsh, cattail marsh, and water, our method achieves 100% accuracy. Based on self-attention, SSTN uses a serial structure that first extracts spatial features and then extracts the spectral feature. Therefore, the spectral module accepts the feature after downscaling the spectral bands of the spatial module, which can lead to distorted features obtained by the spectral module. Our method SSLKA is based on the LKA mechanism, which absorbs the ability of the self-attention mechanism to acquire long-range dependencies. Moreover, it is also able to extract local spatial information. In terms of the network structure, SSLKA instead adopts a parallel structure to extract the spectral–spatial features, which allows simultaneous extraction of undistorted spectral–spatial features. It is clear that our SSLKA method gets better results in terms of classification accuracy.

5) Different Sizes of Training Samples: As shown in Table XII, we conducted experiments on training sets of different sizes to evaluate the classification accuracy of different methods. In particular, we conducted experiments using training samples of 100, 200, 300, and 400. On the IP dataset with a training sample size of 100, our method lags behind the best-performing SSTN by 0.88%. Our method shows higher variance, predominantly because the model does not converge entirely. After gradually increasing the training samples of the IP dataset, our method significantly outperforms other methods in accuracy. The experimental results show that our method achieved the best results on all but one size of the training sets.

6) Running Time Comparison: Table XIII shows the running time of each method on four datasets. From Table XIII, we can see that SSLKA can complete the training process of the model at a fast rate. Compared with 3-D-CNN, SSAN, AUTO, and SSRN, our method trains faster than these methods. This is due to the decomposing of the large kernel convolution and the reduction of the number of training parameters. 3-D-DCNN reduces the number of calculations due to the use of smaller convolutional kernels, thus allowing training to be completed in a short time. SSTN uses a lightweight network structure with a shallow depth of the model, reducing the number of training parameters. This lightweight design allows SSTN to perform relatively well with limited computational resources and complete training in a short time. SDRN achieves the shortest training time by discarding random layers during the training process. Although we used the decomposition of large kernel convolution and reduced the training parameters, we still could not reach the efficiency of SDRN, and we will continue to lighten our network structure in the future. Additionally, our approach outperforms other methods in terms of classification accuracy.

D. Ablation Studies

To further demonstrate the impact of each part of our proposed SSLKA, we performed ablation experiments while keeping the other experimental settings the same. Since our proposed method is on the basis of LKA and LKA can be decomposed to DW-Conv, DW-D-Conv, PW-Conv, we removed each part of DW-Conv, DW-D-Conv, PW-Conv, and LKA, that is, without (w/o) DW-Conv, (w/o) DW-D-Conv, (w/o) PW-Conv, and (w/o) LKA, to test the corresponding performances. To determine the extent to which the spatial attention branch and spectral attention branch influence the results of the experiments, we similarly set up experiments without (w/o) the spatial attention branch and without (w/o) the spectral attention branch. Table XIV shows the results of the ablation studies performed on the IP, HT, UP, and KSC datasets, OA, AA, and kappa. The experimental results prove that DW-Conv, DW-D-Conv, and PW-Conv are positive for the enhancement of the LKA module. Meanwhile, the spatial attention branch and the spectral attention branch play significant roles in our model. Our proposed method makes better

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use of local information, captures long-range dependencies, and is adaptive in the channel dimension.

V. Conclusion

In this article, we introduced an SSLKA for HSI classification. Different from the current feature extraction method, our proposed method uses the large convolutional kernel attention mechanism to better extract the global relation and local features. Moreover, several consecutive cooperative spectral–spatial feature extraction blocks are also designed to explore features in spectral and spatial domains. The experimental results show that our proposed method leads to a significant performance improvement. In the future, we will explore more attention mechanisms focusing on global relations and local features to further improve the classification performance.

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